



CS & IT ENGINEERING

Discrete Mathematics

Mathematical logic

Lecture No.- 07



By- Satish Sir

Recap of Previous Lecture



Topic

Logic Part-06



Topics to be Covered



Topic

Logic Part-07



$$\neg \forall x A(x) \equiv \exists x \neg A(x)$$

$$\neg \exists x B(x) \equiv \forall x \neg B(x)$$

negate the logical expression.

$$1) \quad \forall x [p(x) \rightarrow q(x)]$$

negate:

$$\neg \forall x [p(x) \rightarrow q(x)]$$

$$\neg \forall x [\neg p(x) \vee q(x)]$$

$$\exists x \neg [\neg p(x) \vee q(x)]$$

$$\exists x [p(x) \wedge \neg q(x)]$$

negate it:

$$2) \quad \exists x [A(x) \vee B(x)]$$

$$\neg \exists x [A(x) \vee B(x)]$$

$$\forall x [\neg A(x) \wedge \neg B(x)]$$

$$\neg \forall x A(x)$$

$$\exists x \neg A(x)$$

negation
of predicate.

Doubt:

$$\exists x A(\neg x)$$

~~X~~

negation
of subject

$\neg A(x)$: x is not even no.

$A(x)$: x is even no.

$A(y)$: y is even no.

$A(z)$: z is even no.

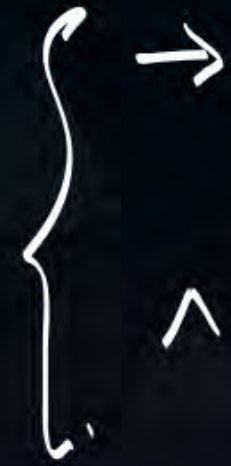
$A(\neg x)$: $\neg x$ is even no.

All mothers are female.



All of x , x is mother x is female.
 $\forall x$ $MT(x)$ $F(x)$

$\forall x [MT(x) \rightarrow F(x)]$



All mothers are female.

D:

Case I:

$$\forall x [m_T(x) \rightarrow F(x)]$$

T.	False	→		T.
		∧		
	False	→		T.
		∧		
	T	→	T	T.

Case II:

$$\forall x [m_T(x) \wedge F(x)]$$

<u>False</u>	False	∧		F.
		∧		
	False	∧		F.
		∧		
	T	∧	T	T.

- Satisf.
- single female.
- mT/F.

ALL →

* All comedians are funny.

$$\forall x [C(x) \rightarrow F(x)]$$

* All graphs are connected.

$$\forall x [G(x) \rightarrow C(x)]$$

Some cats are black.



Some of x , x is cat x is black.
 $\exists x$ $CT(x)$ $BL(x)$

$\exists x [CT(x) \wedge BL(x)]$

Case I:

$$\exists x [CT(x) \rightarrow BL(x)]$$

T	False $\rightarrow$	True.
	$\vee$	
	$\rightarrow$	
	$\vee$	
	$\rightarrow$	

Case II

$$\exists x [CT(x) \wedge BL(x)]$$

T	False $\wedge$	F.
	$\vee$	
	F $\wedge$	F.
	$\vee$	
	T $\wedge$ T	T.

Some cats are black.

$\downarrow$ D: Animal.

- cow
- Dog
- CAT/BL.
-

Case I:

$$\exists x [CT(x) \rightarrow BL(x)]$$

T	False. $\rightarrow$	T
	$\vee$	
	$\rightarrow$	
	$\vee$	
	$\rightarrow$	

Case II

$$\exists x [CT(x) \wedge BL(x)]$$

False	F	$\wedge$	F
	$\vee$		
	F	$\wedge$	F
	$\vee$		
	T	$\wedge$	F

Some cats are black.

False. $\downarrow$ D: Animal.

- cow
- Dog
- CAT/w

Some real no. are rational. (GATE)

$$\rightarrow \exists x [\text{Real}(x) \wedge \text{Rational}(x)]$$

$\left\{ \begin{array}{l} \underline{\text{ALL}} \rightarrow \\ \underline{\text{some}} \wedge \end{array} \right.$

1. All Graphs are connected. 2. not (all Graphs are connected)

$$\forall x [G(x) \rightarrow C(x)]$$

$$\neg \forall x [G(x) \rightarrow C(x)]$$

$$2 \neq 4.$$

$$3 = 4.$$

3. All Graphs are disconnected. $\equiv$ 4. no Graphs are connected.

$$\forall x [G(x) \rightarrow \neg C(x)]$$

$$\forall x [G(x) \rightarrow \neg C(x)]$$

eg:

not all rainy days are cold.

$$\neg \forall x [\text{Rainy}(x) \rightarrow \text{Cold}(x)]$$

no mothers are female

$$\forall x [\text{mother}(x) \rightarrow \neg f(x)]$$

not all $\neg \forall x [\rightarrow]$

no/none. $\forall x [\rightarrow \neg]$

eg:

none of my friends are perfect.

$$\forall x [F(x) \rightarrow \neg P(x)]$$

Type-4.

↓
box method:

Some cats are black.

$\left\{ \begin{array}{l} \text{True} \\ \text{False} \end{array} \right.$

I:	II:
$\wedge$	$\rightarrow$
$\begin{bmatrix} T \\ F \end{bmatrix}$	$\begin{bmatrix} T \\ T \end{bmatrix}$

Some $\rightarrow \wedge$



2 mins Summary

Topic

One

Logic Part-07

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU